CNENGR202 Instrumentation and Control

Part 3 L7 – PWM - II

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Recap

- We saw how to create a frequency generator
- Now real PWM

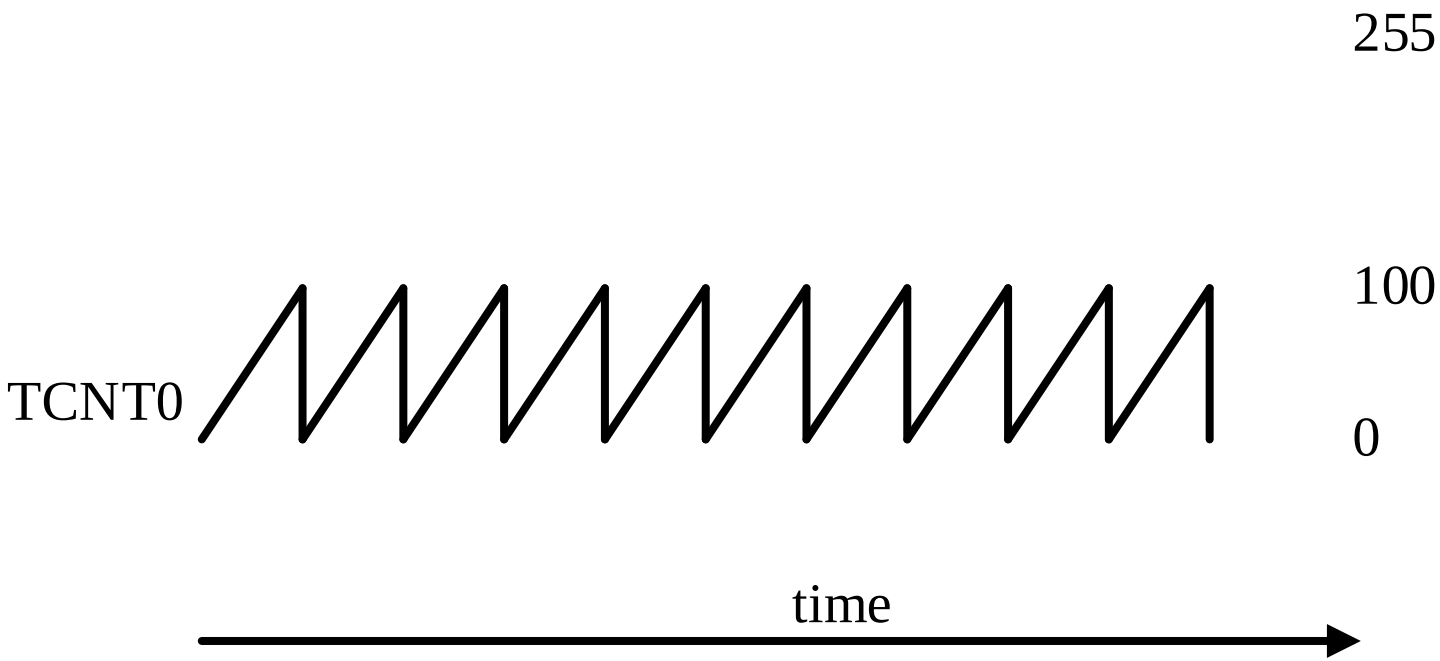
Waveform Generation

- First we shall look at Fast PWM

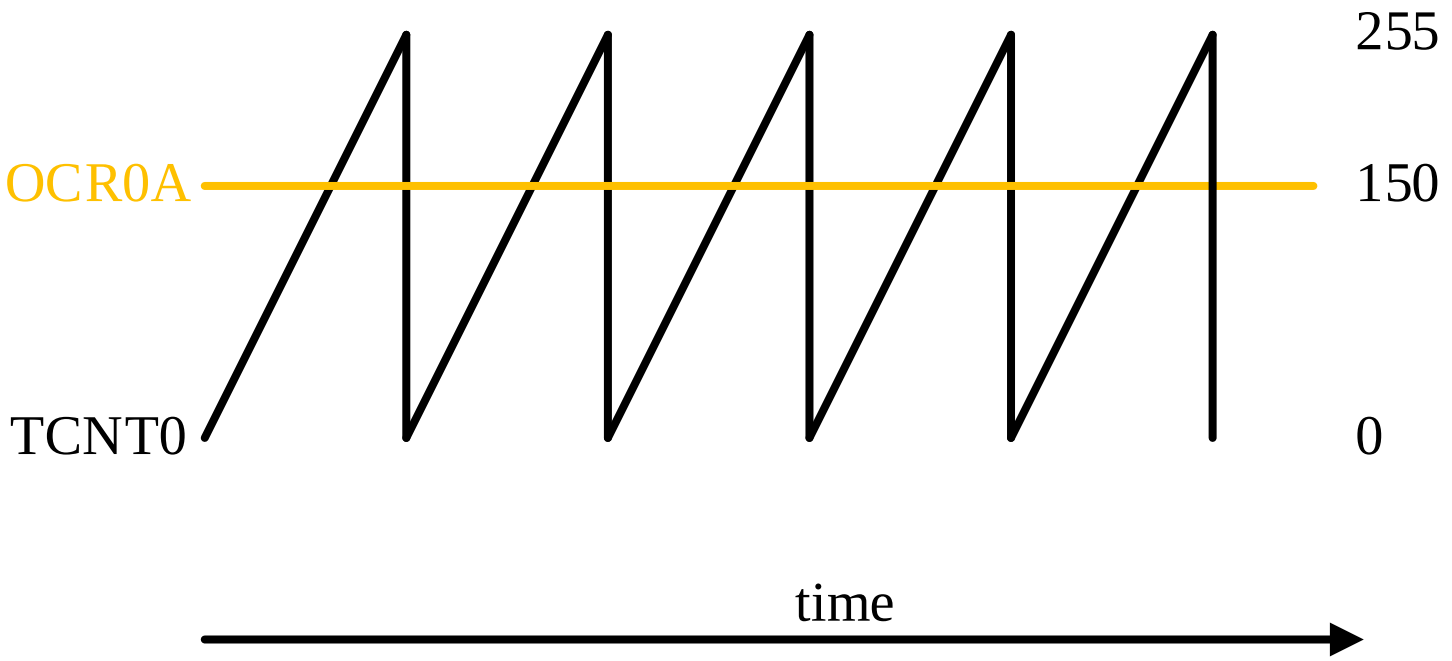
Waveform Generation Mode Description

Mode	WGM 02	WGM 01	WGM 00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on
0	0	0	0	Normal	0xFF	immediate	MAX
1	0	0	1	PWM, Phase correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	-	-	-
5	1	0	1	PWM, Phase correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	-	-	-
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

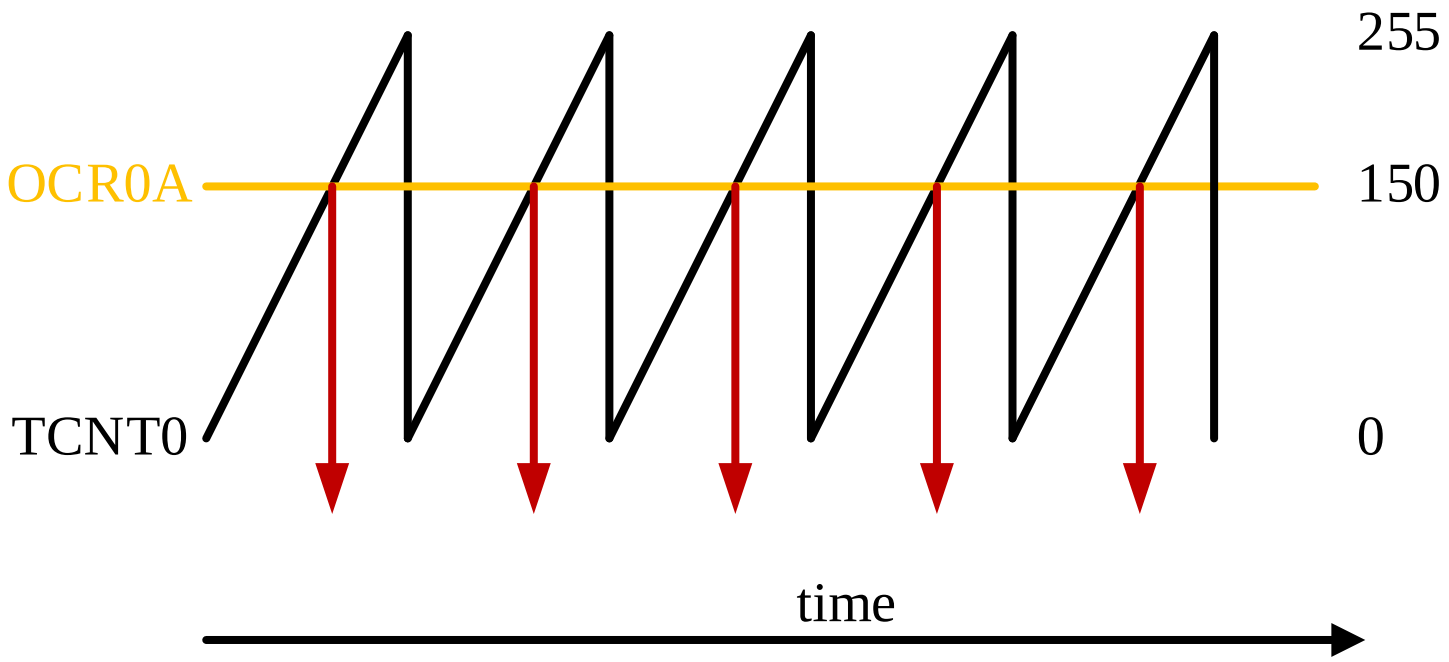
Timing Diagram - I



Timing Diagram - II

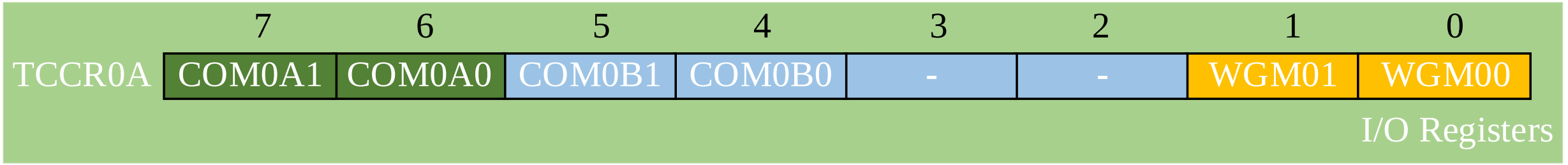


Timing Diagram - III



Timer Counter Control Register A

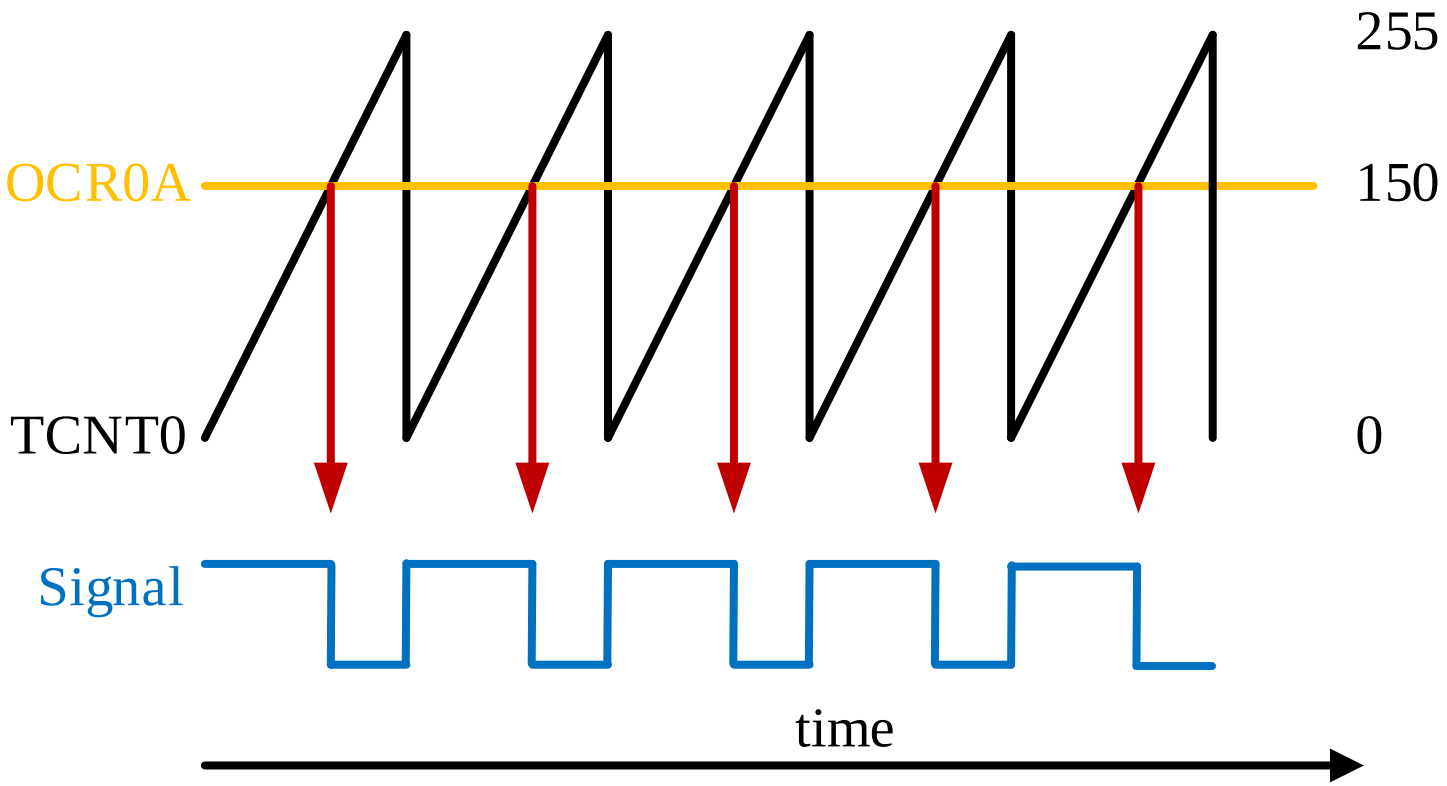
- Set up differently than for CTC mode



Compare Output Mode, Fast PWM Mode

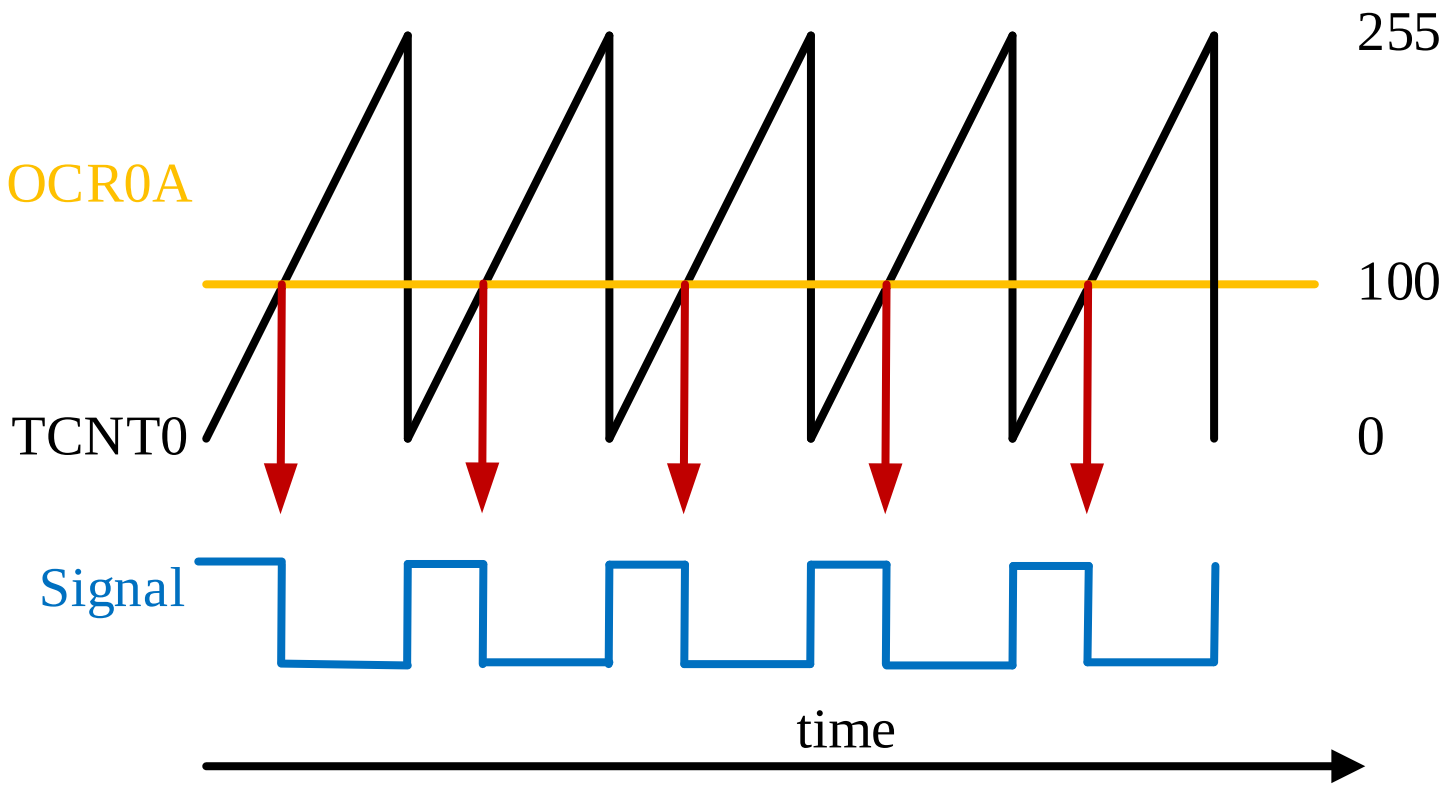
COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected
0	1	WGM02=0: Normal Port Operation, OC0A disconnected WGM02=1: Toggle OC0A on Compare Match
1	0	Clear OC0A on Compare Match, set OC0A at BOTTOM, (non-inverting mode).
1	1	Set OC0A on Compare Match, clear OC0A at BOTTOM, (inverting mode).

Timing Diagram - I



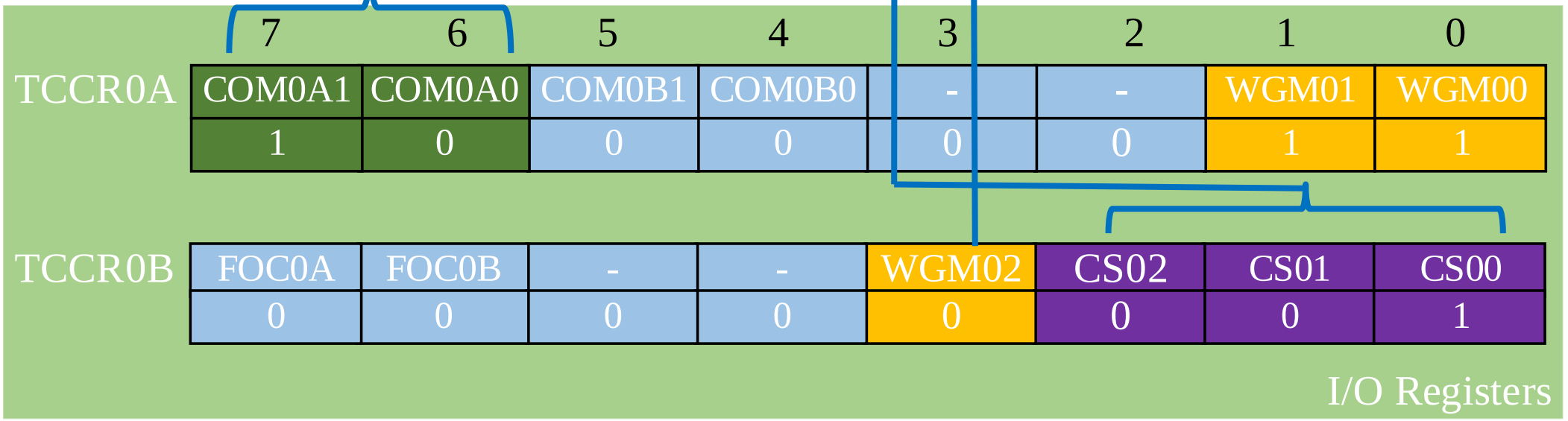
Clear output on match, set on bottom

Timing Diagram - II



Clear timer on compare match

- A very simple timer mode
- Resets the counter when it reaches the value of OCR0A
- Same operation we have just reviewed
- WGM0n = 011 (Fast PWM mode)
- COM0An = 10 (clear on match, set on bottom)
- CS02 = 001 (/1)



Fast PWM settings

- The OC0A PWM output is PortD pin 6.

DDRD	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0

- Set the point at which the output will change

  OCR0A = 100; // Set the compare register

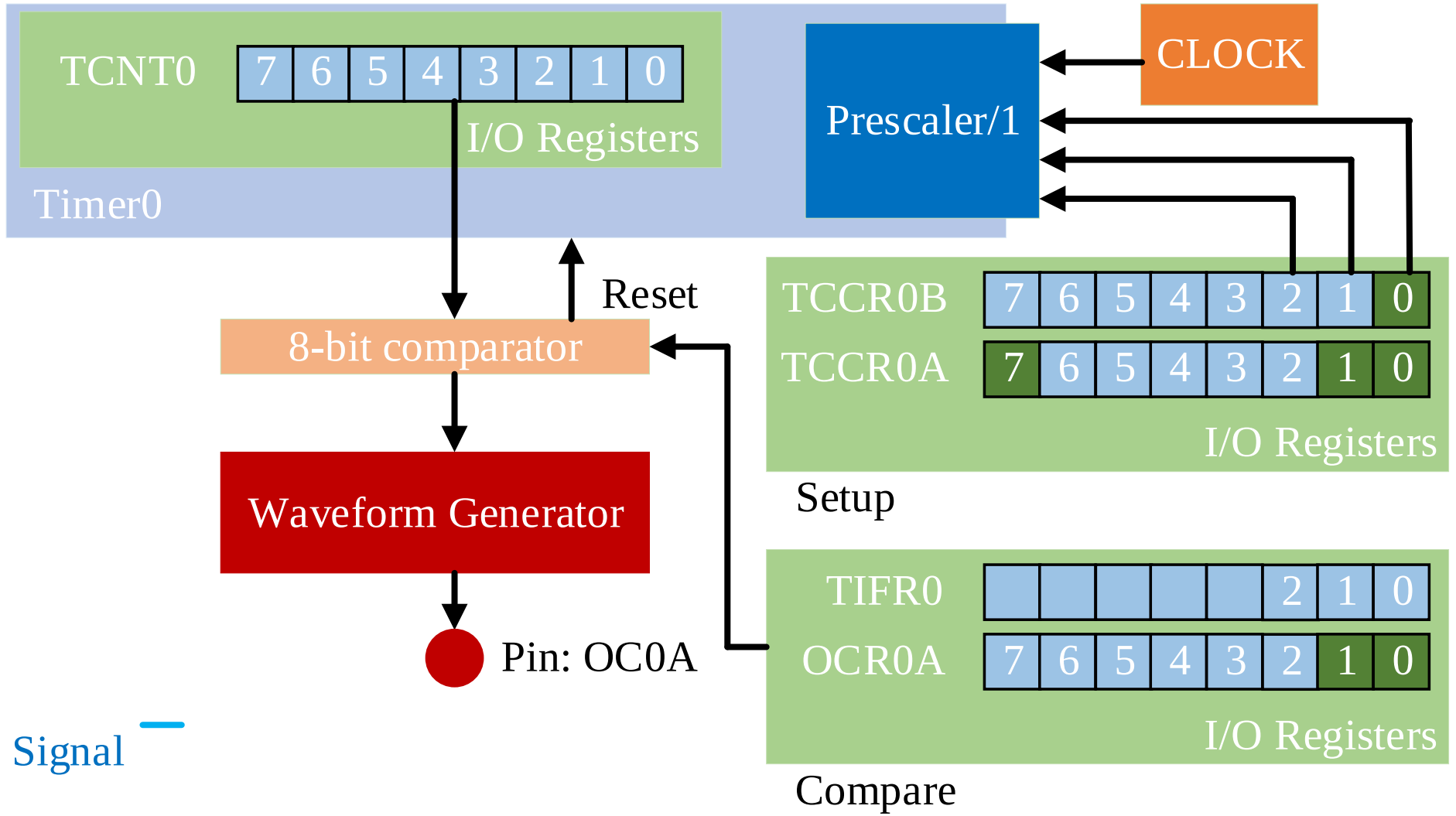
- Sets COM0A1, WGM01 & WGM00 for non inverting fast PWM

  TCCR0A = 0b1000_0011; // Setup the control register A

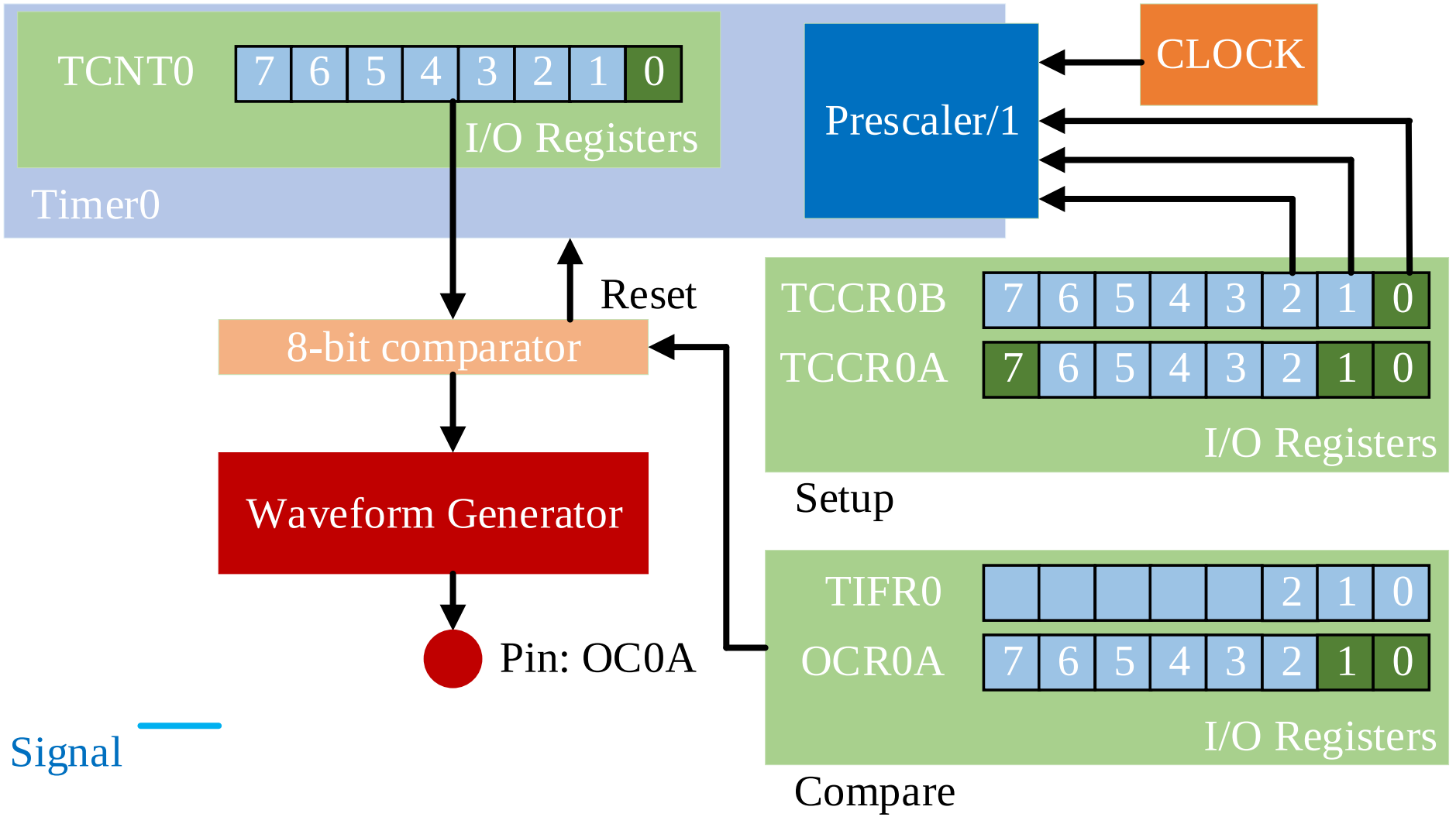
- Sets up the prescaler (see lecture 5)

  TCCR0B = 0b0000_0001; //Setup the control register A

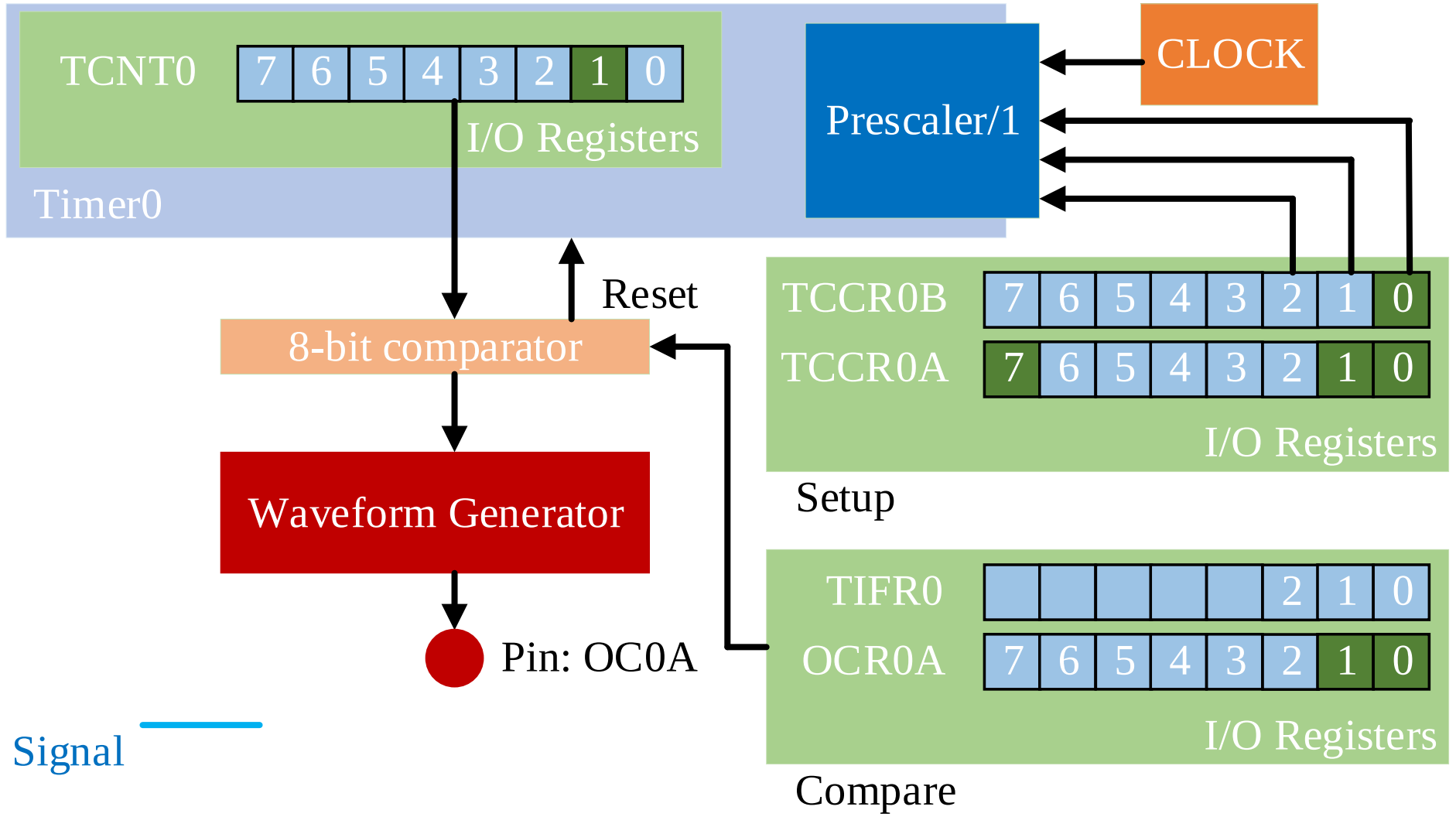
Frequency Control - I



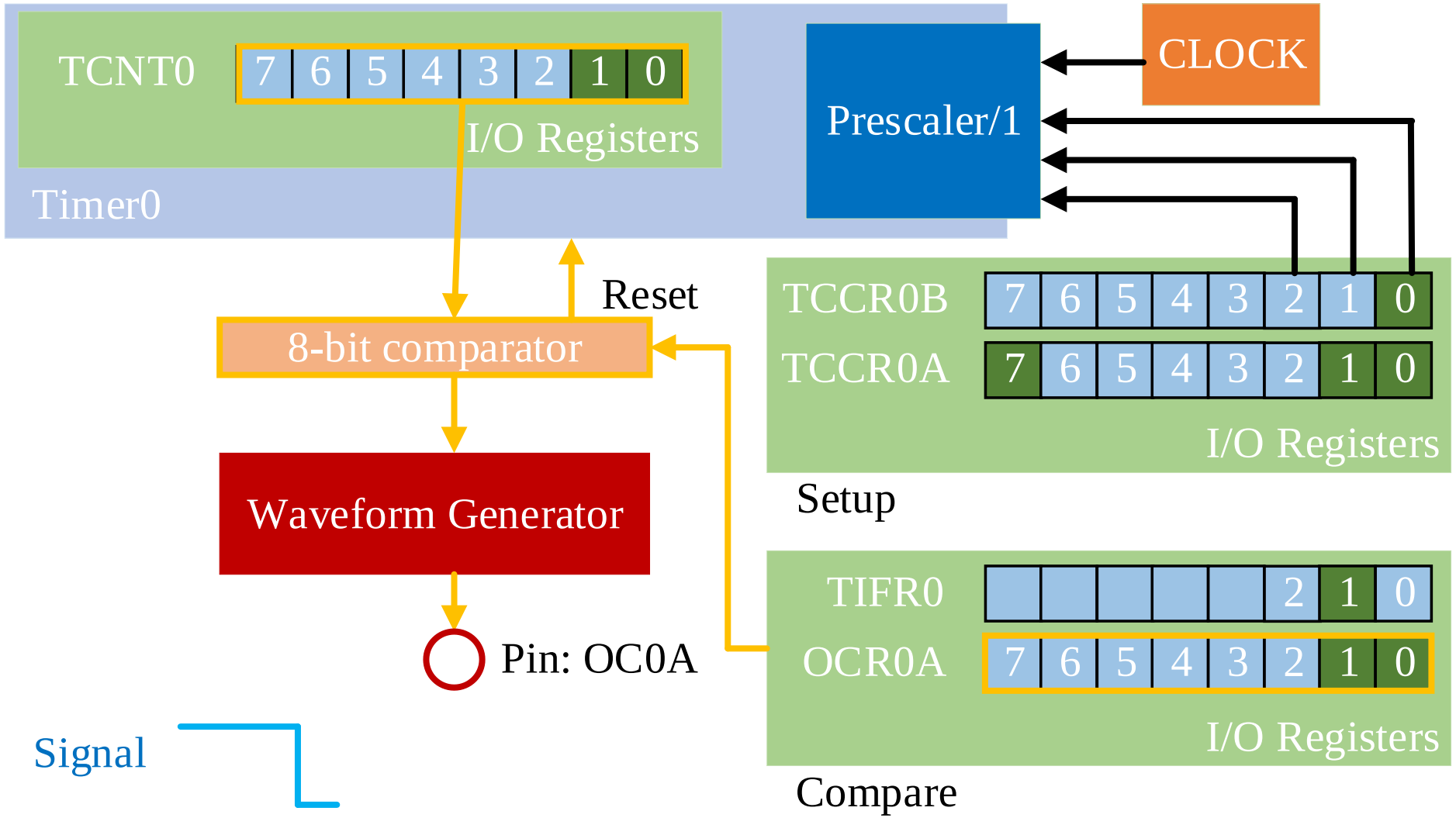
Frequency Control - II



Frequency Control - III



Frequency Control - IV

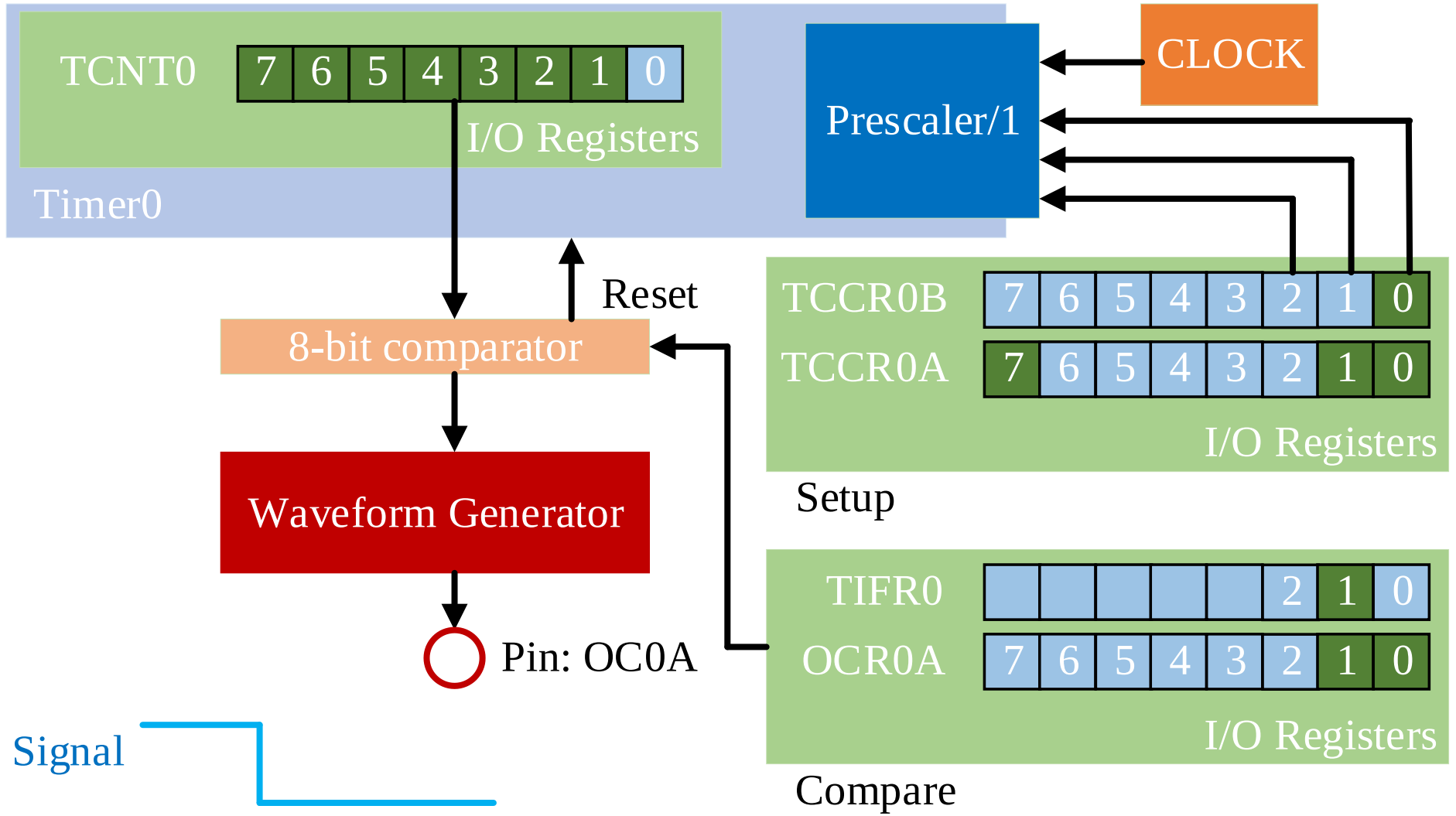


Frequency Control - V

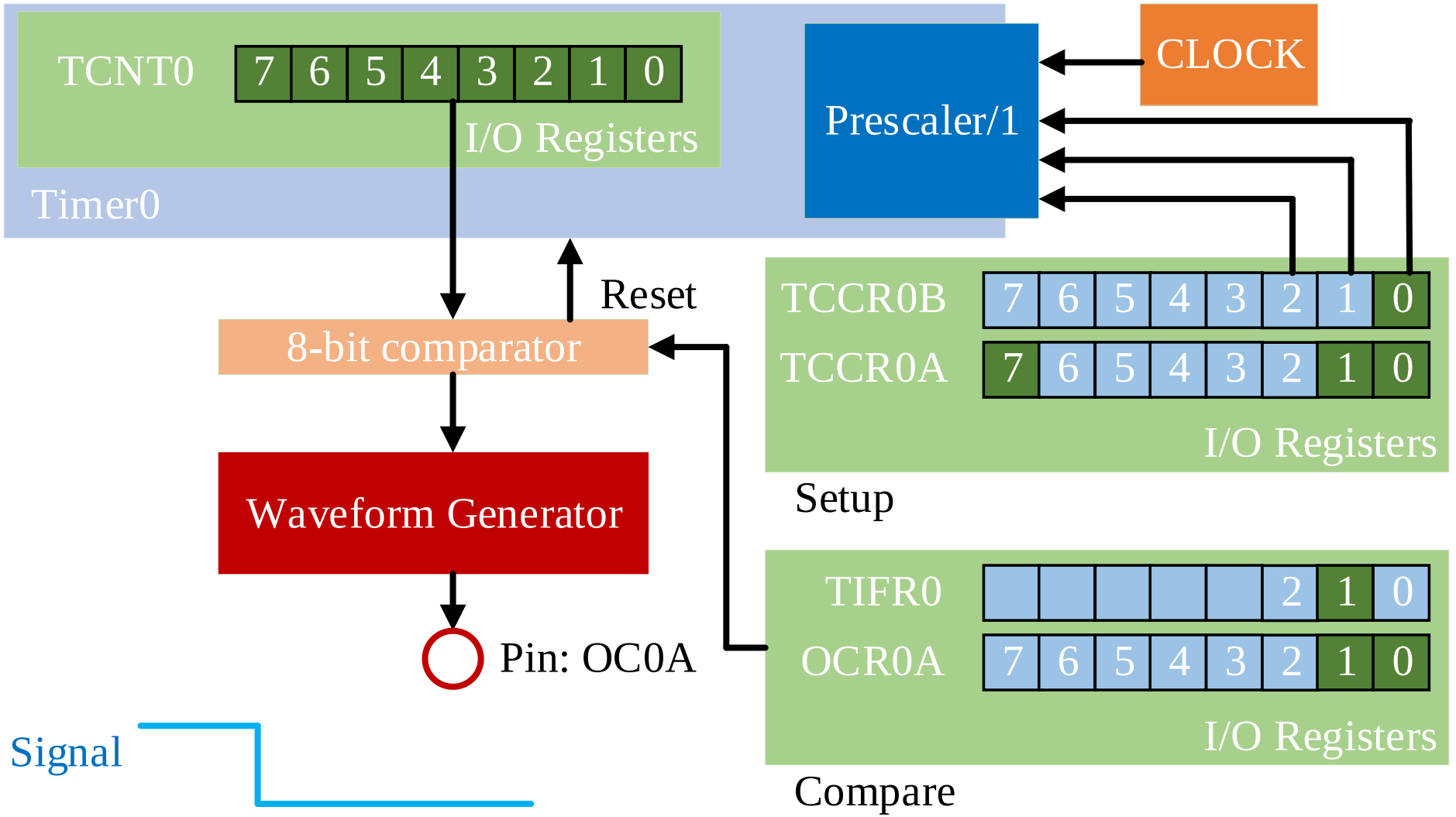
Some time later



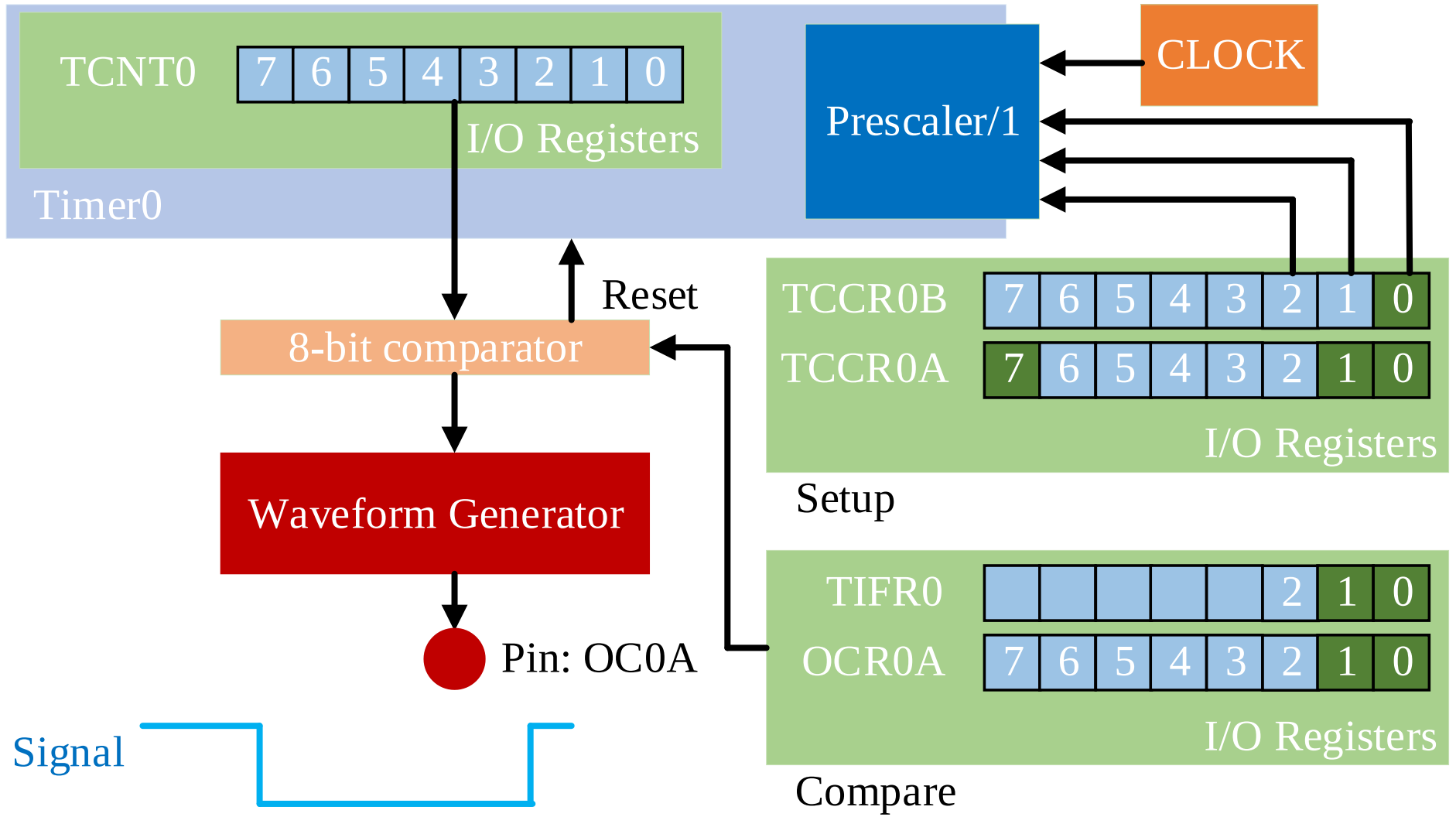
Frequency Control - VI



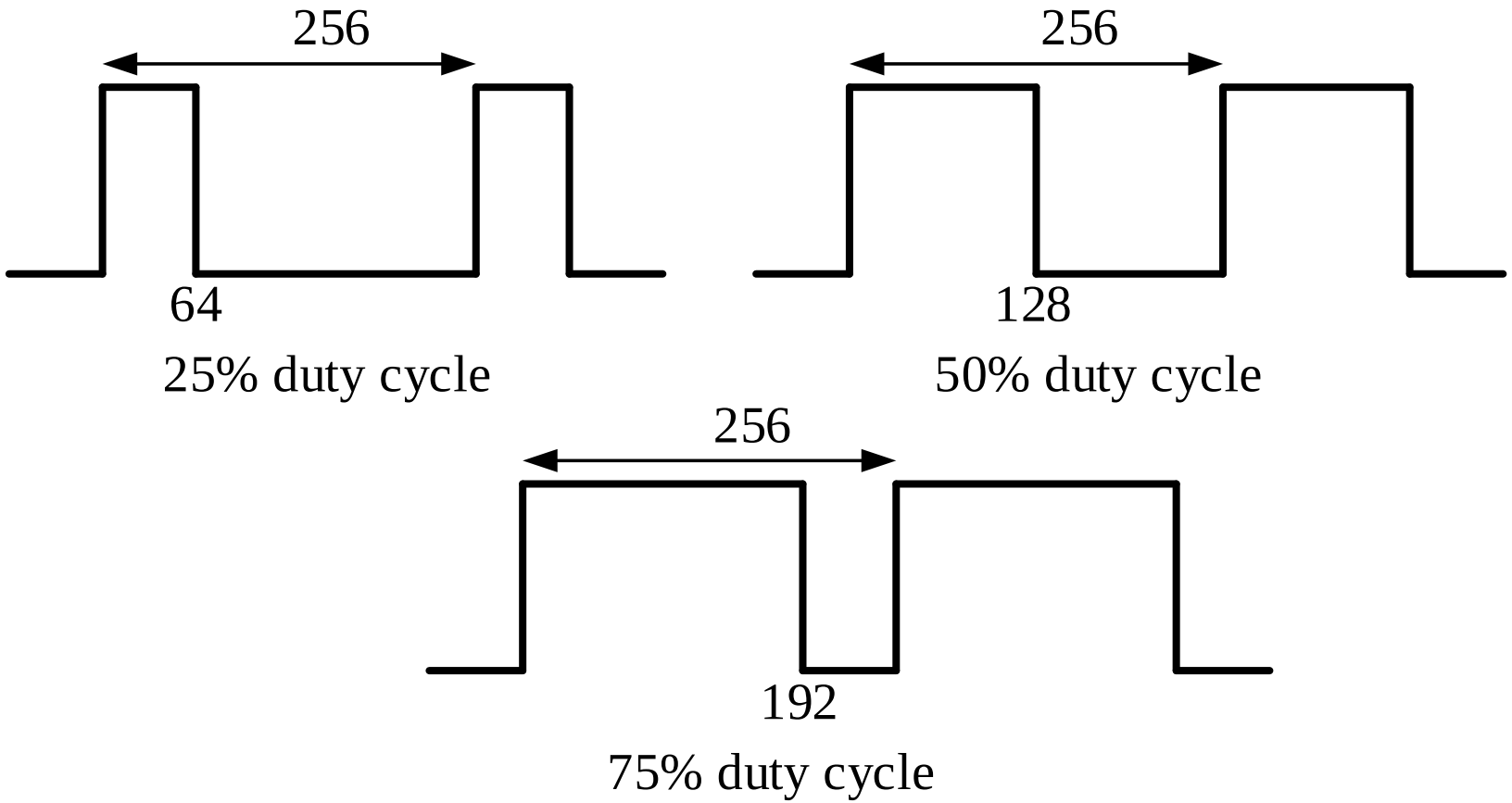
Frequency Control - VII



Frequency Control - VIII



PWM duty cycle comparisons



Phase Correct PWM

Waveform generation

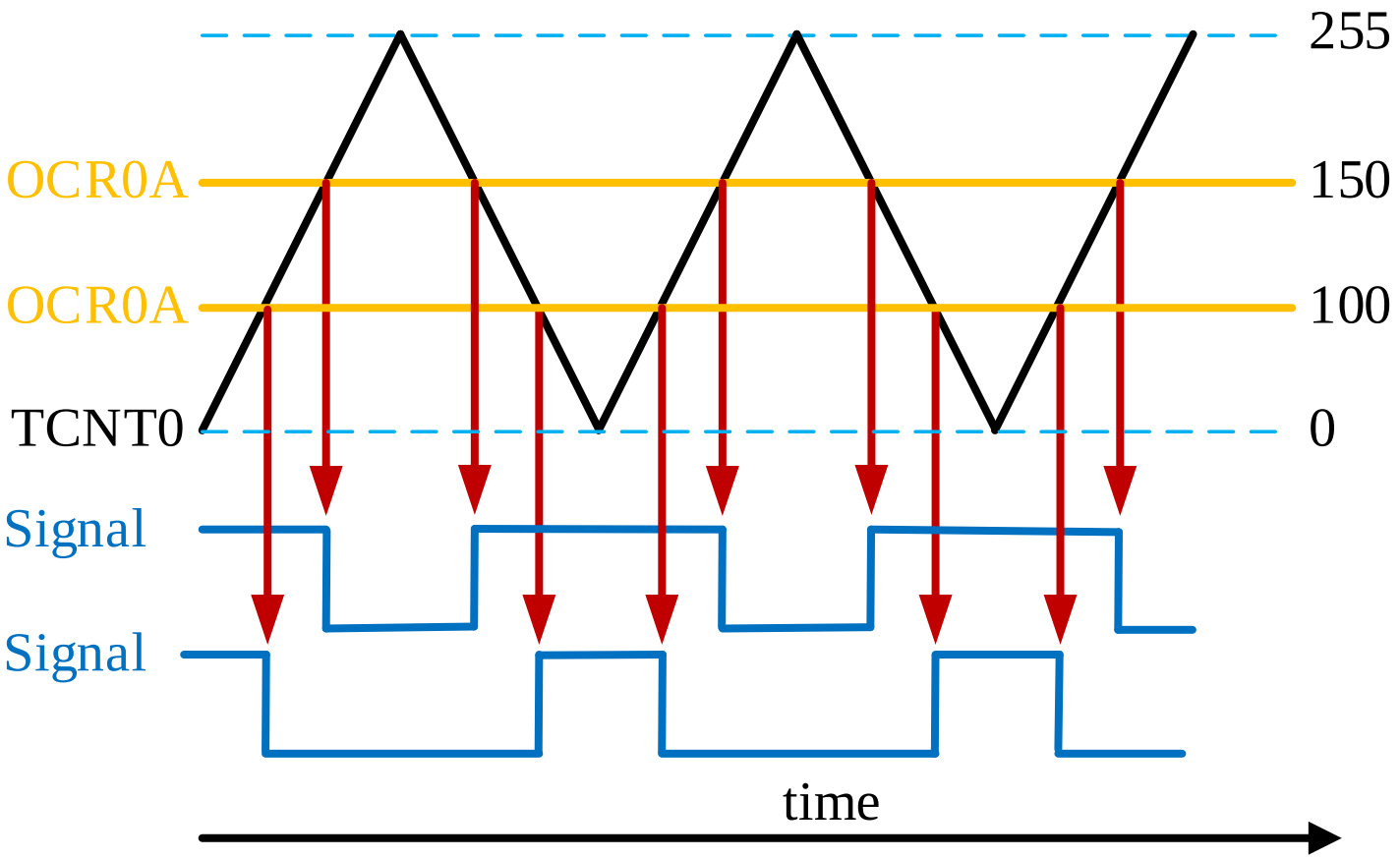
Waveform generation Mode Bit Description

Mode	WGM 02	WGM 01	WGM 00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on
0	0	0	0	Normal	0xFF	immediate	MAX
1	0	0	1	PWM, Phase correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	-	-	-
5	1	0	1	PWM, Phase correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	-	-	-
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

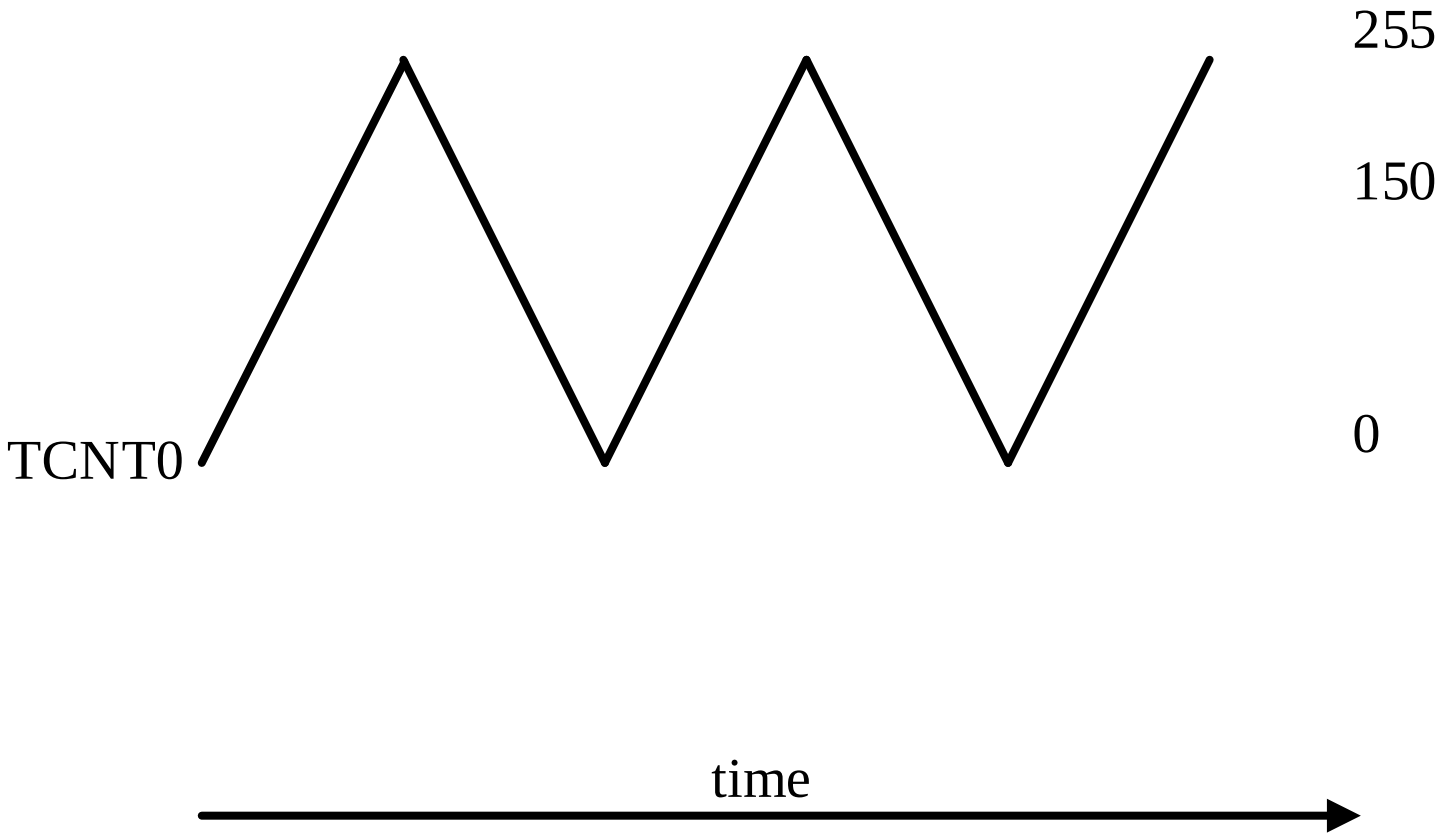
Phase correct PMW

- Counts up to 255 then back down to 0 again.
- Output is low when counter > OCR0A, otherwise on
- Symmetrical output, keeps the centre of the pules in phase
- Half frequency because timer goes up and down

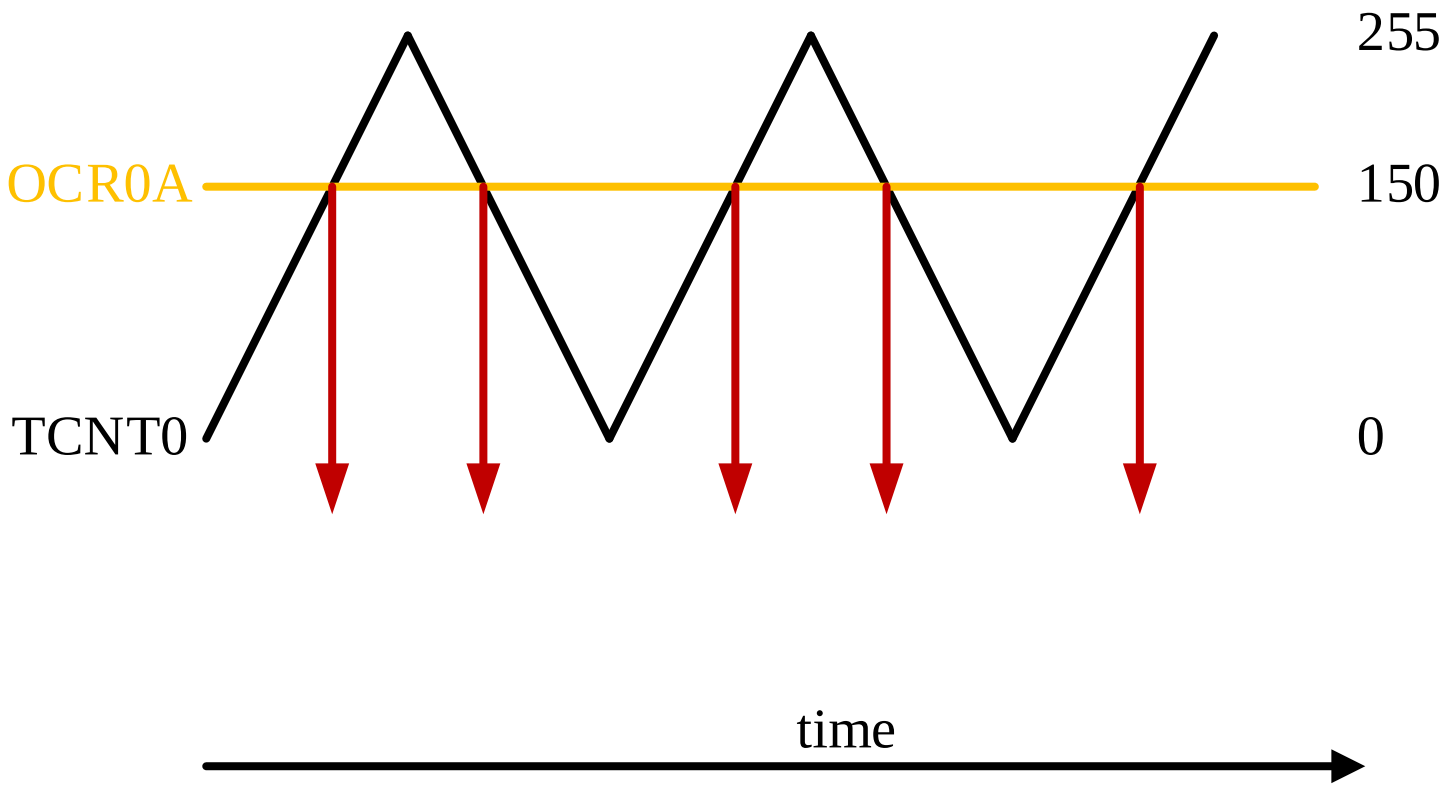
Timing Diagram (Fast PWM)



Timing Diagram (Phase Correct PWM)



Timing Diagram (Phase Correct PWM)



Timer Counter Control Register A

- Controls the pin overrides.
  - 2 bit COM0An controls the connection of OC0A to it's pin (12)

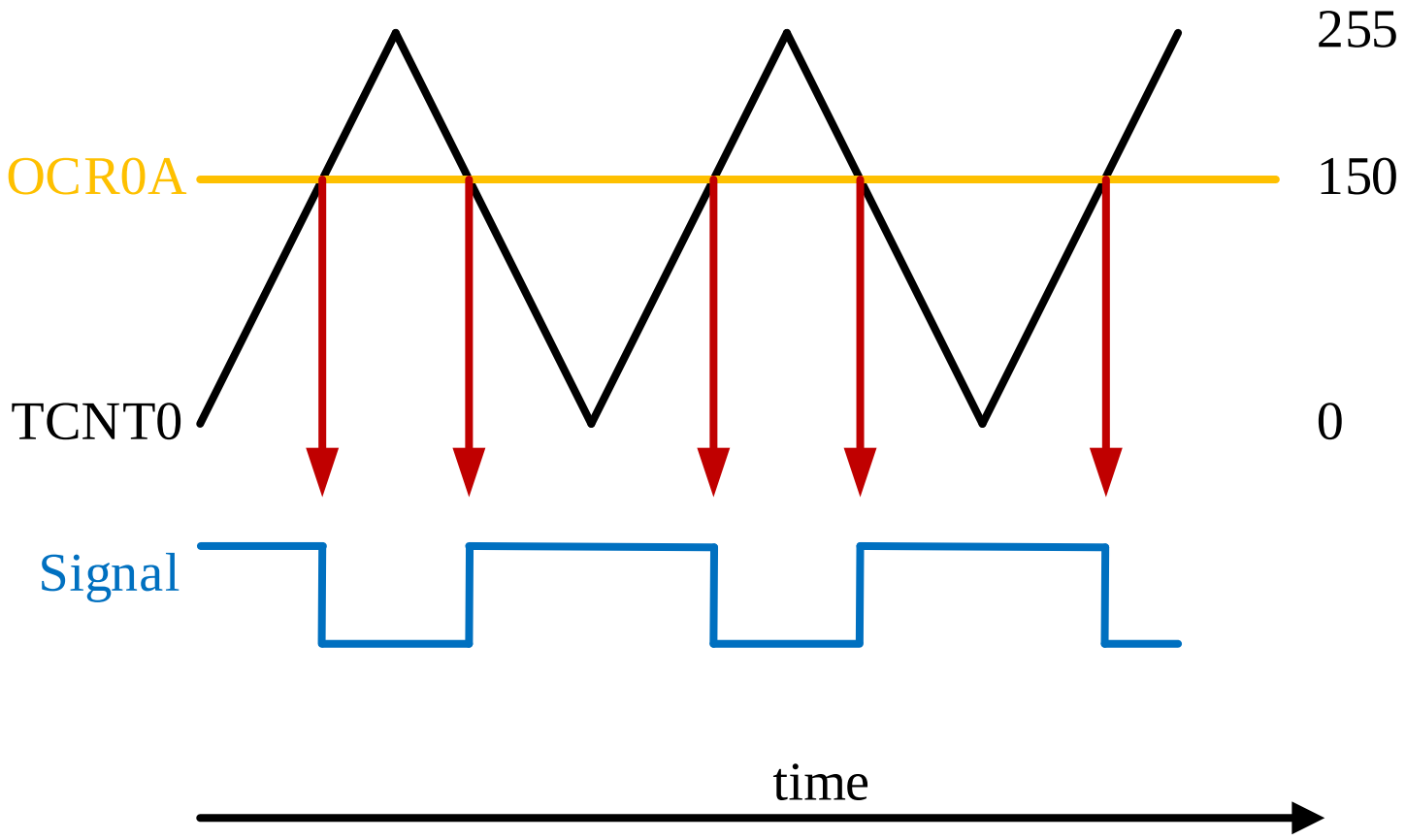
	7	6	5	4	3	2	1	0
TCCR0A	COM0A1	COM0A0	COM0B1	COM0B0	-	-	WGM01	WGM00

I/O Registers

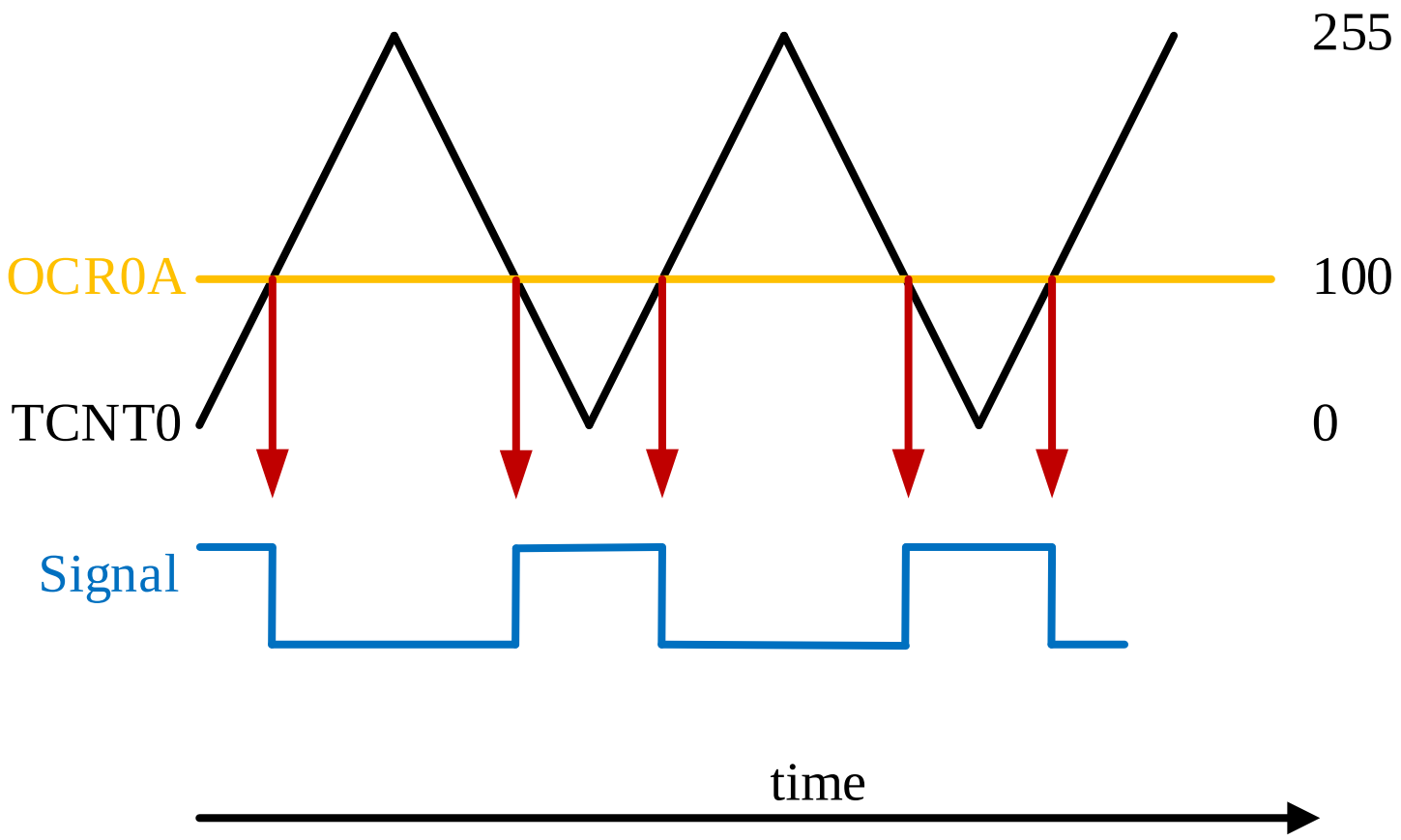
Compare Output Mode, Fast PWM Mode

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected
0	1	WGM02=0: Normal Port Operation, OC0A disconnected WGM02=1: Toggle OC0A on Compare Match
1	0	Clear OC0A on Compare Match, set OC0A at BOTTOM, (non-inverting mode).
1	1	Set OC0A on Compare Match, clear OC0A at BOTTOM, (inverting mode).

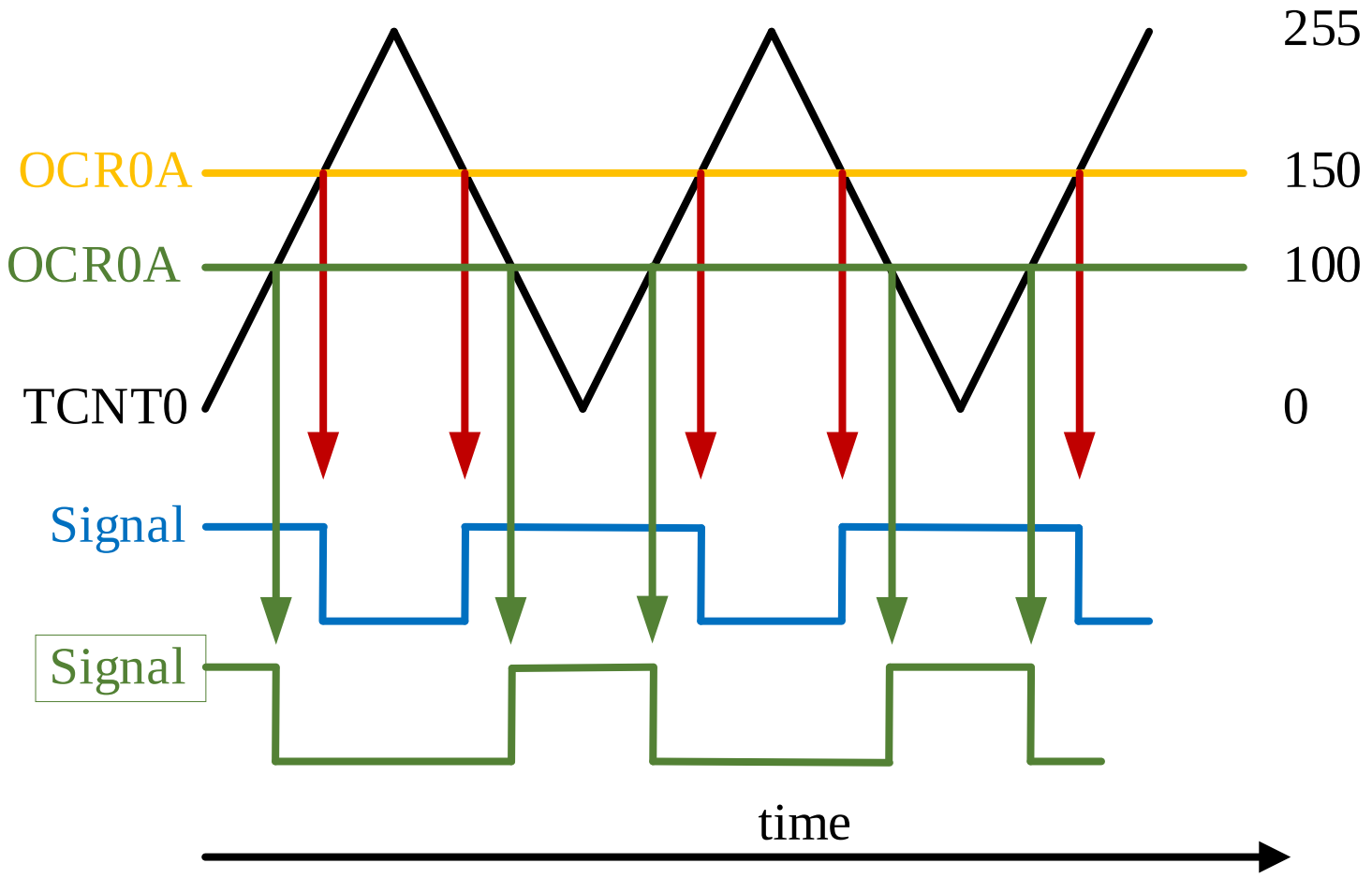
Timing Diagram (Phase Correct PWM)



Timing Diagram (Phase Correct PWM)

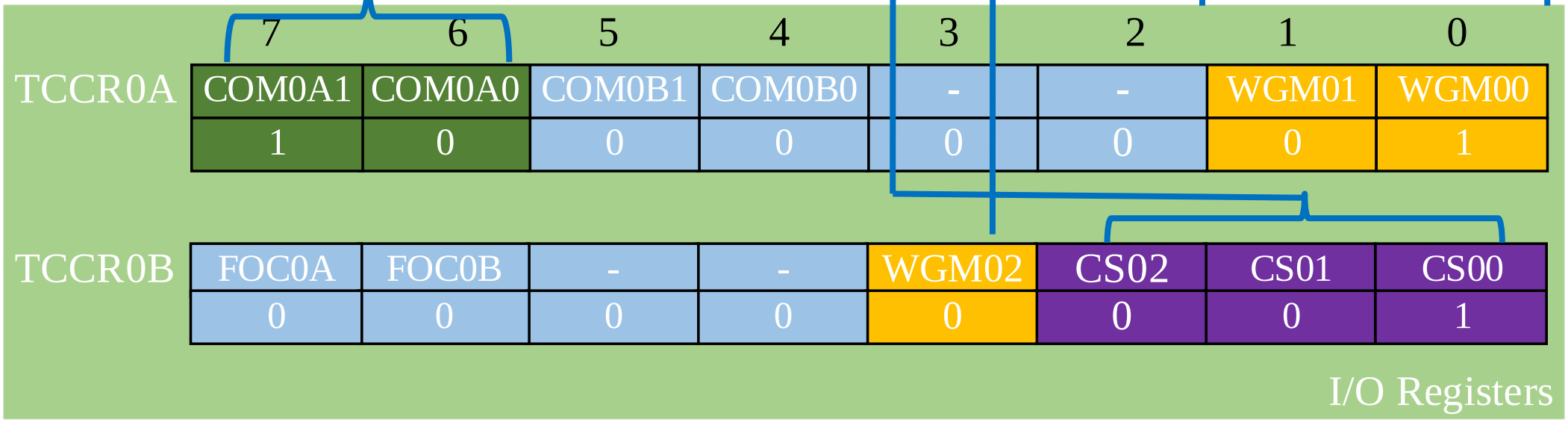


Timing Diagram (Phase Correct PWM)

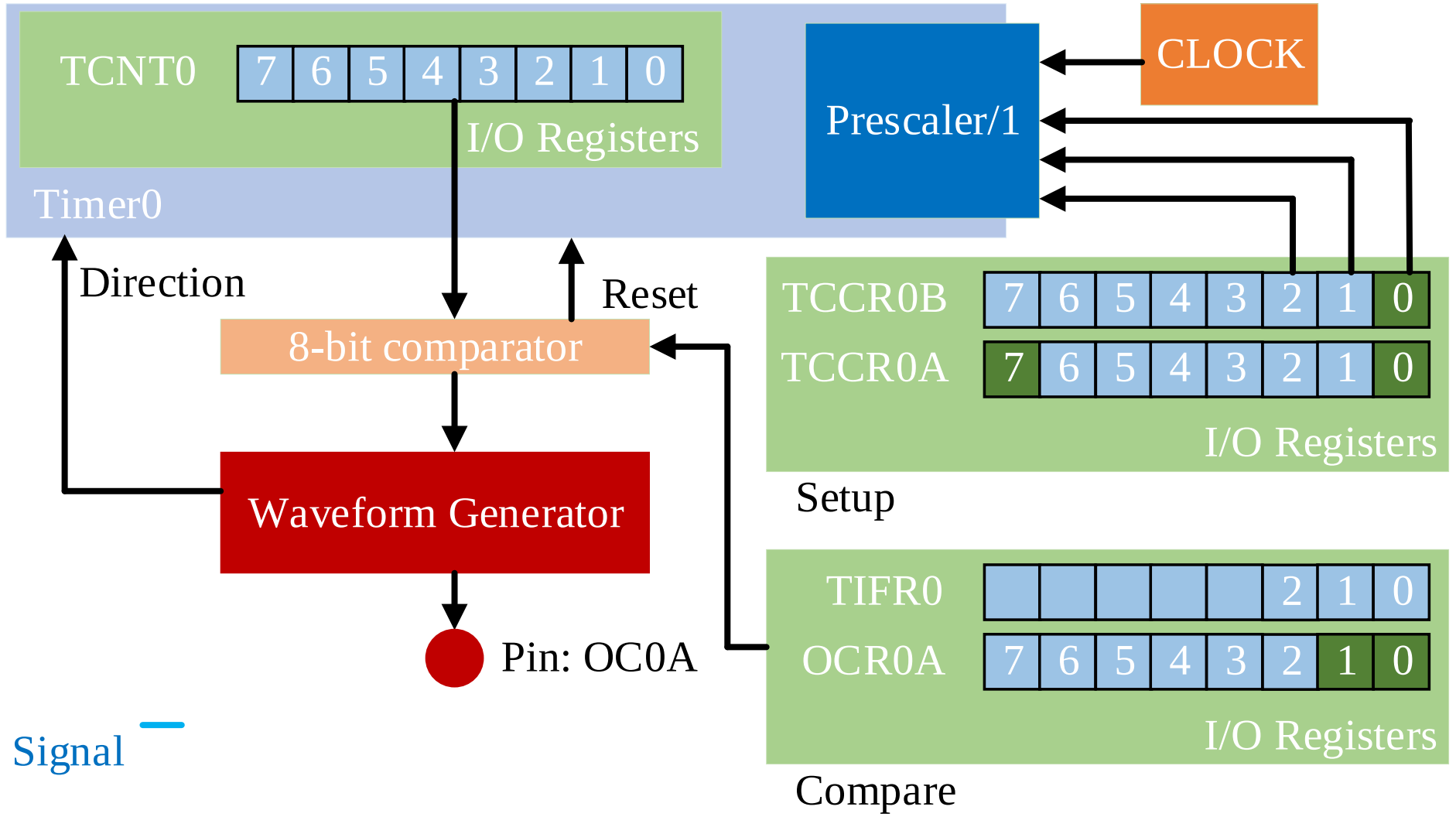


Clear timer on compare match

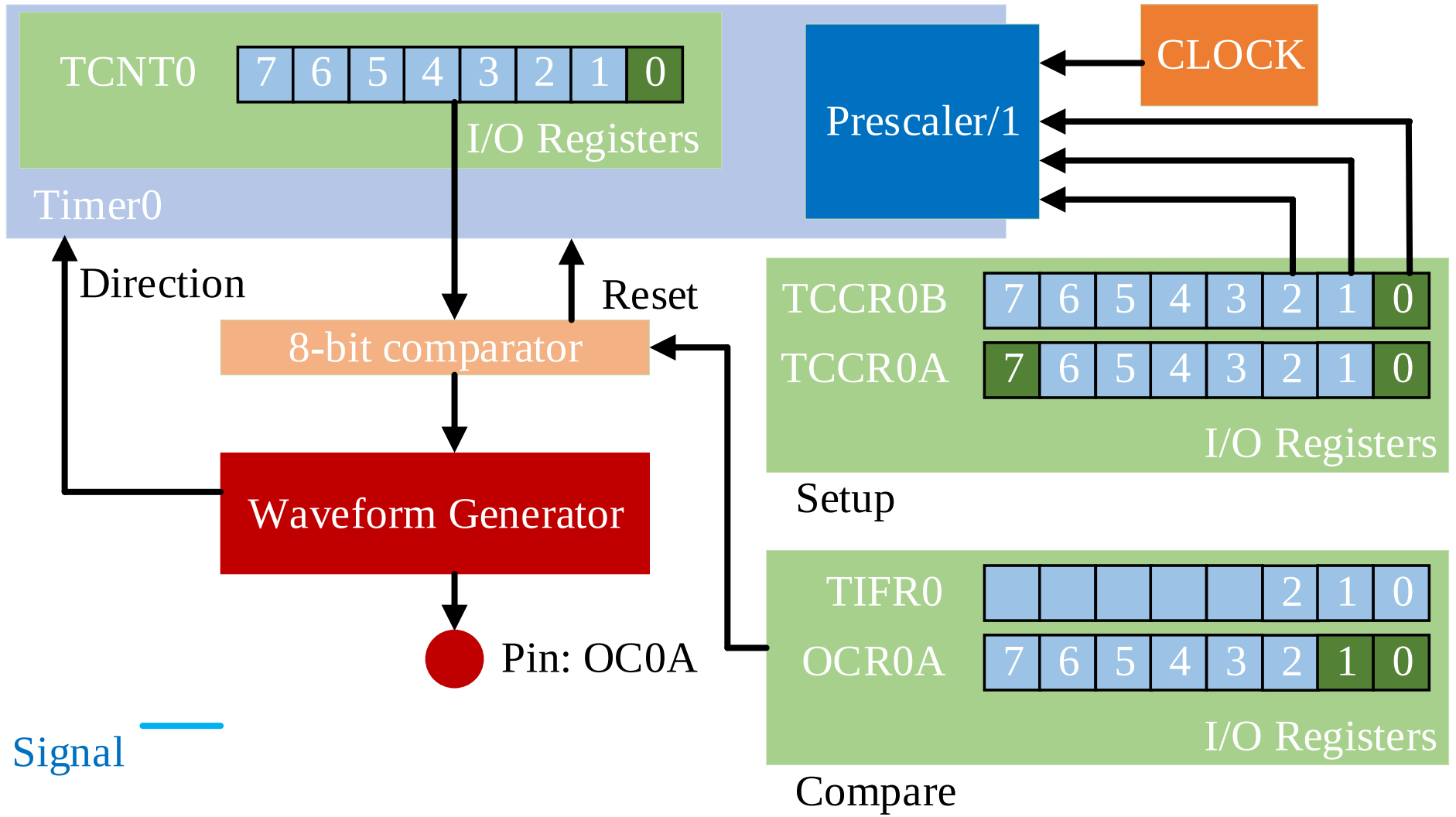
- A very simple timer mode
- Resets the counter when it reaches the value of OCR0A
- Same operation we have just reviewed
- WGM0n = 001 (Phase correct mode)
- COM0An = 10 (clear on match, set on down bottom)
- CS02 = 001 (/1)



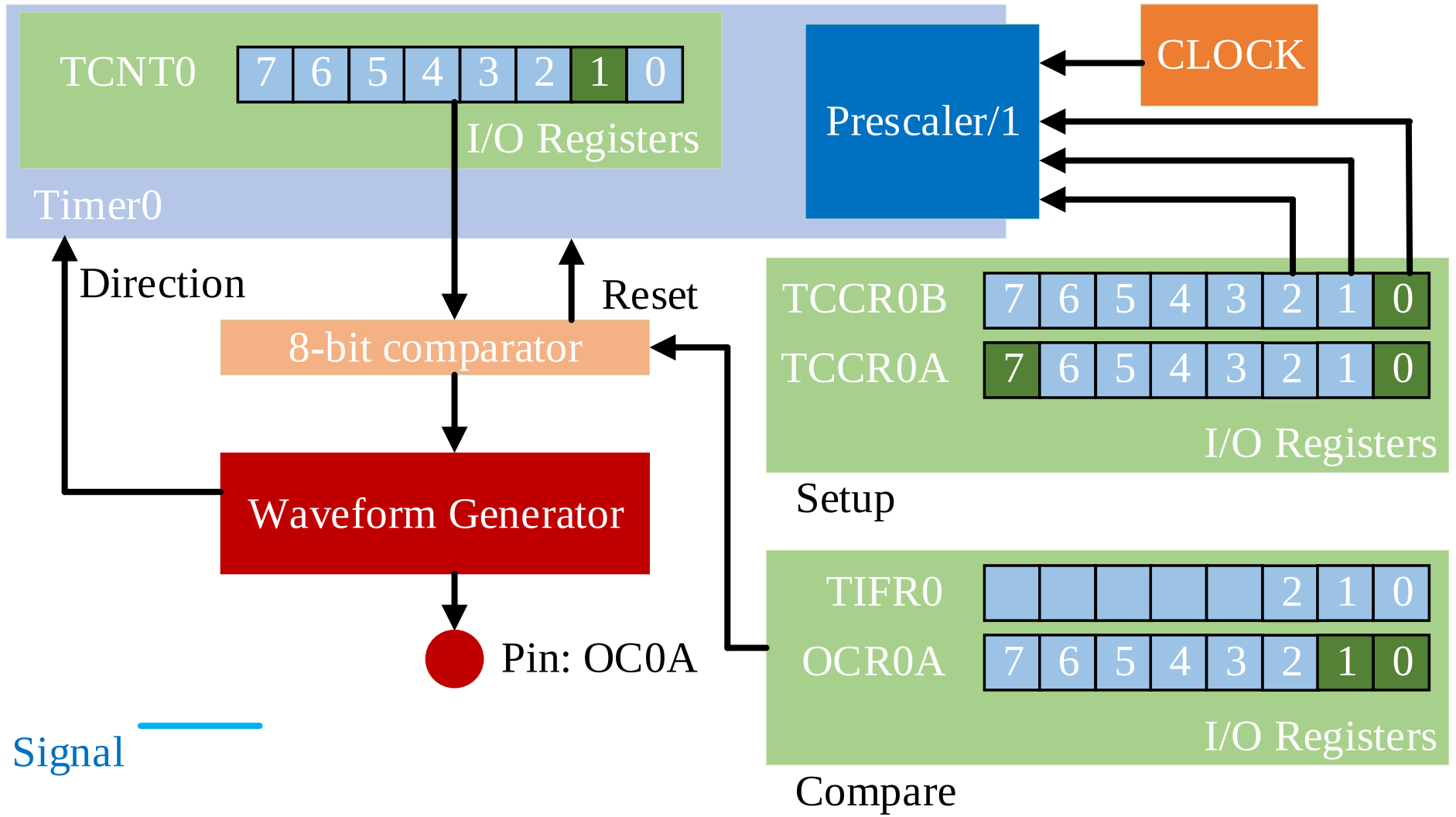
Phase Correct PWM - I



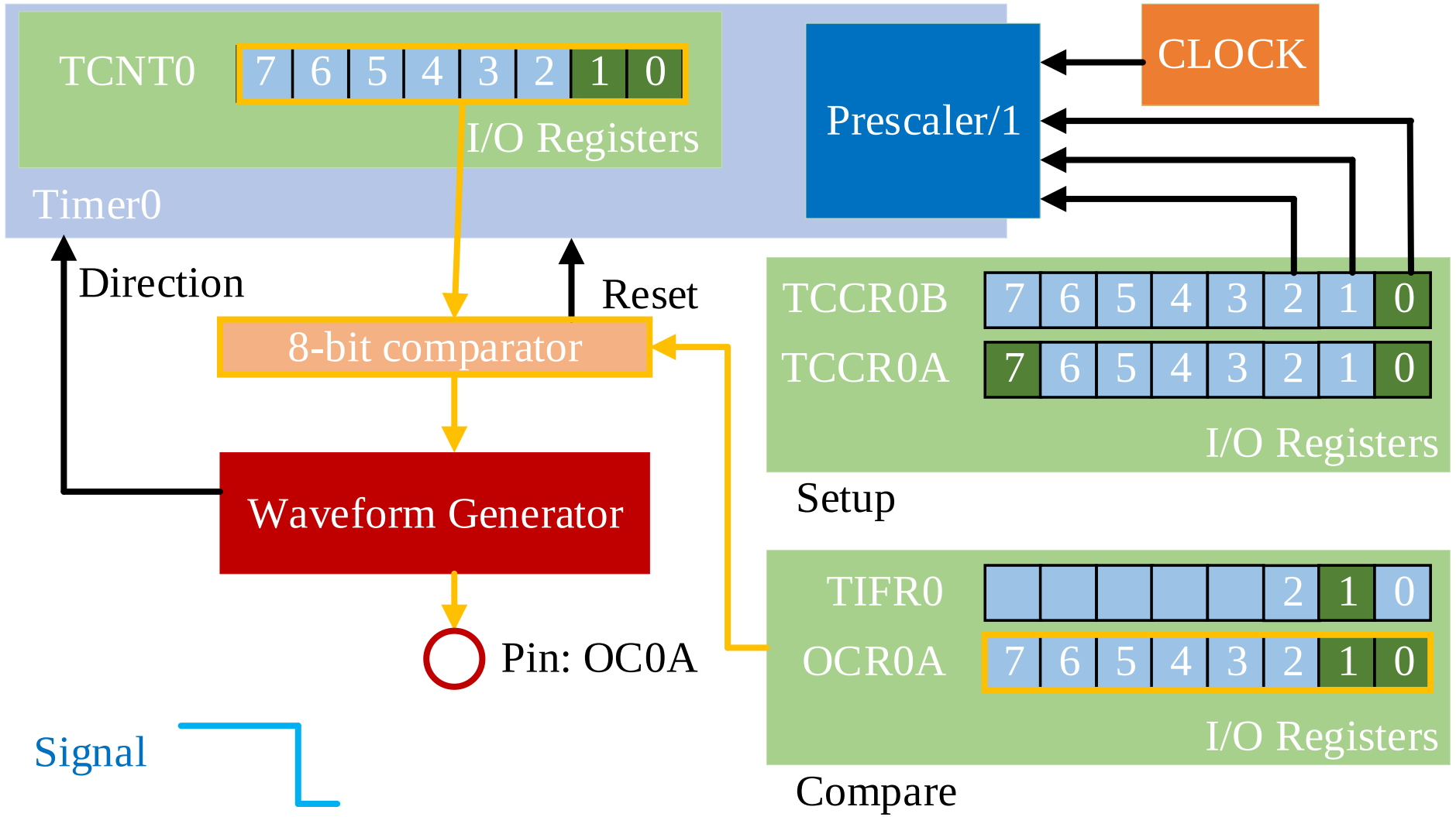
Phase Correct PWM - II



Phase Correct PWM - III



Phase Correct PWM - IV

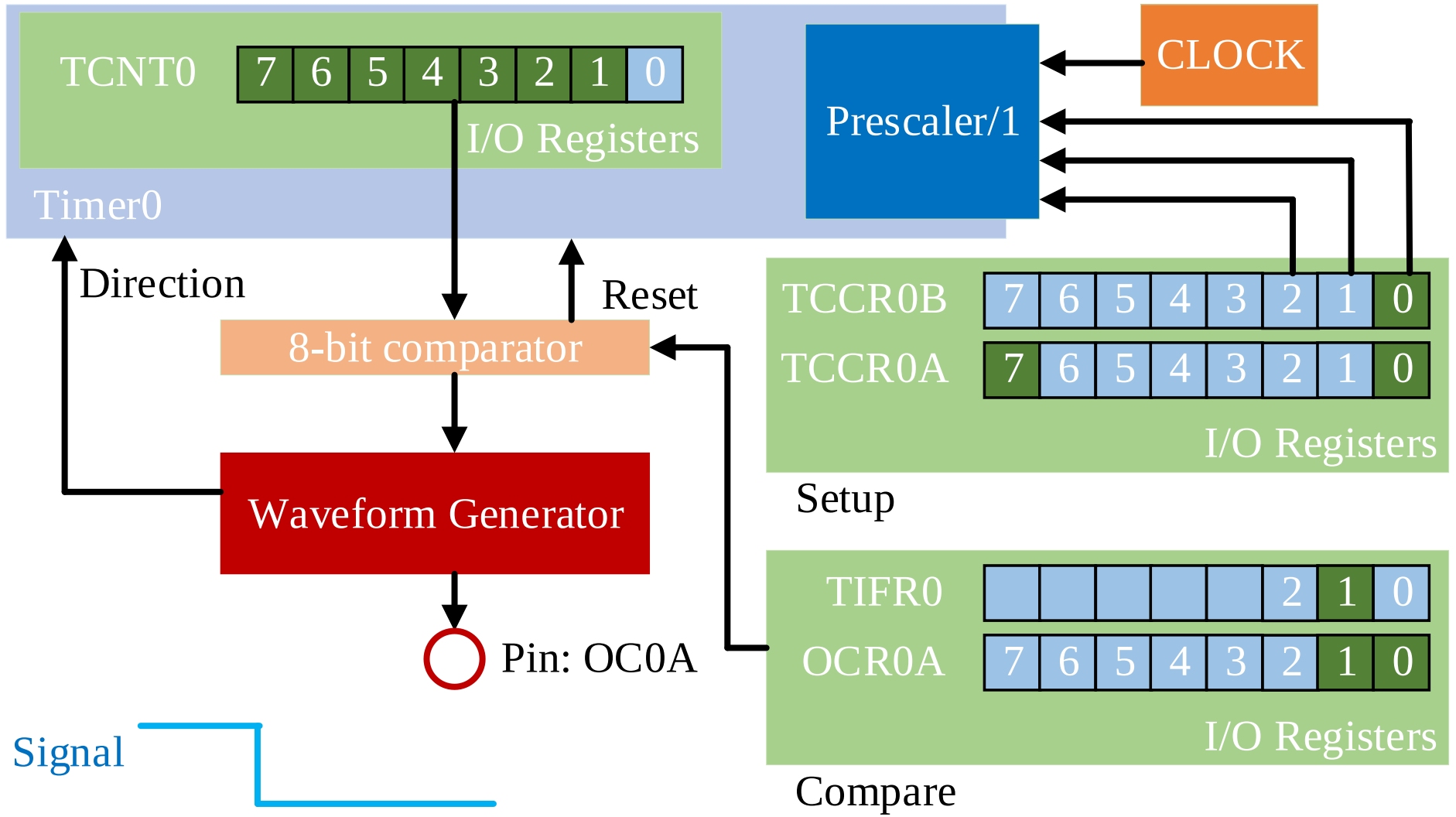


Phase Correct PWM - V

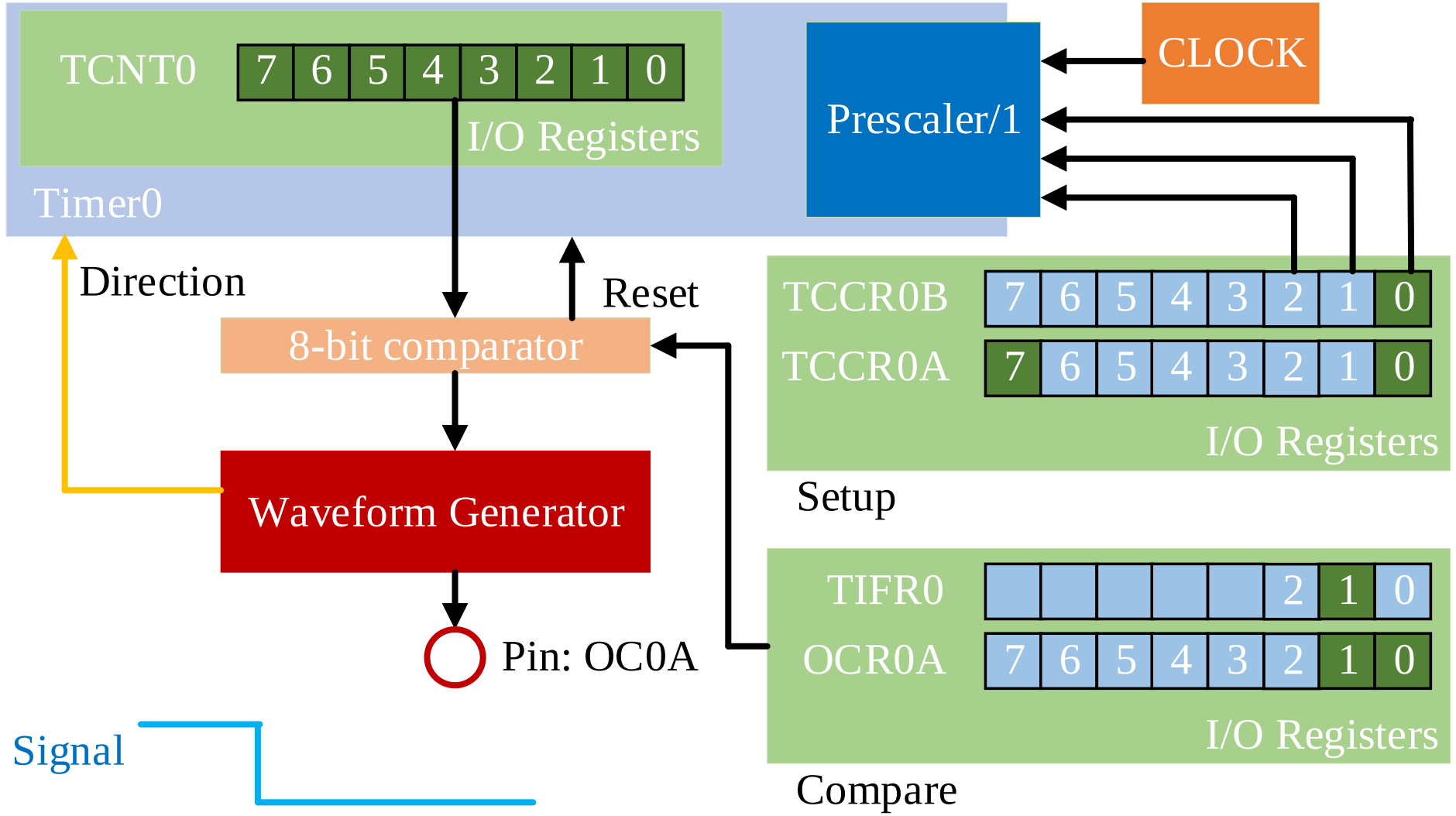
Some time later



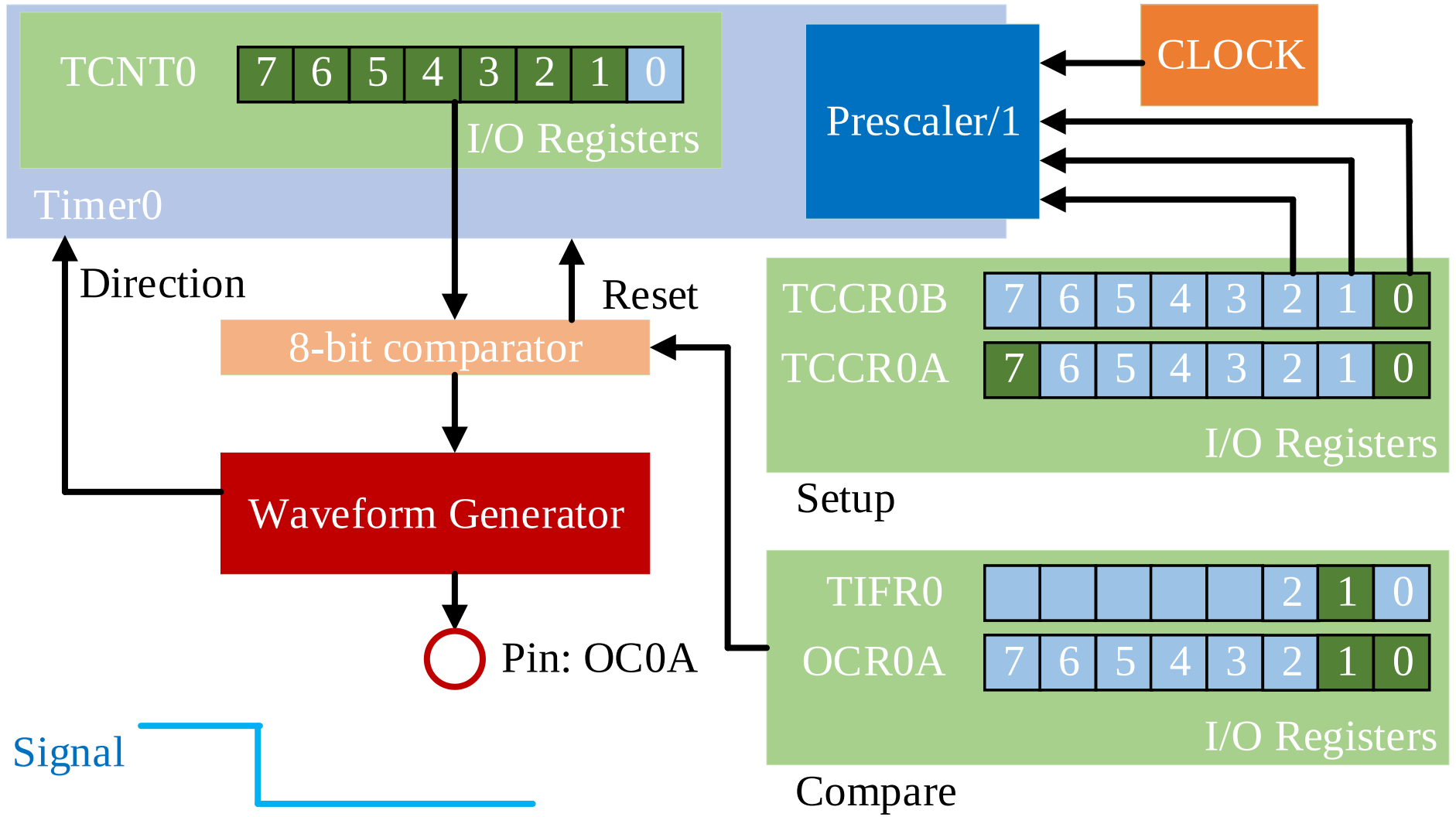
Phase Correct PWM - VI



Phase Correct PWM - VII



Phase Correct PWM - VIII

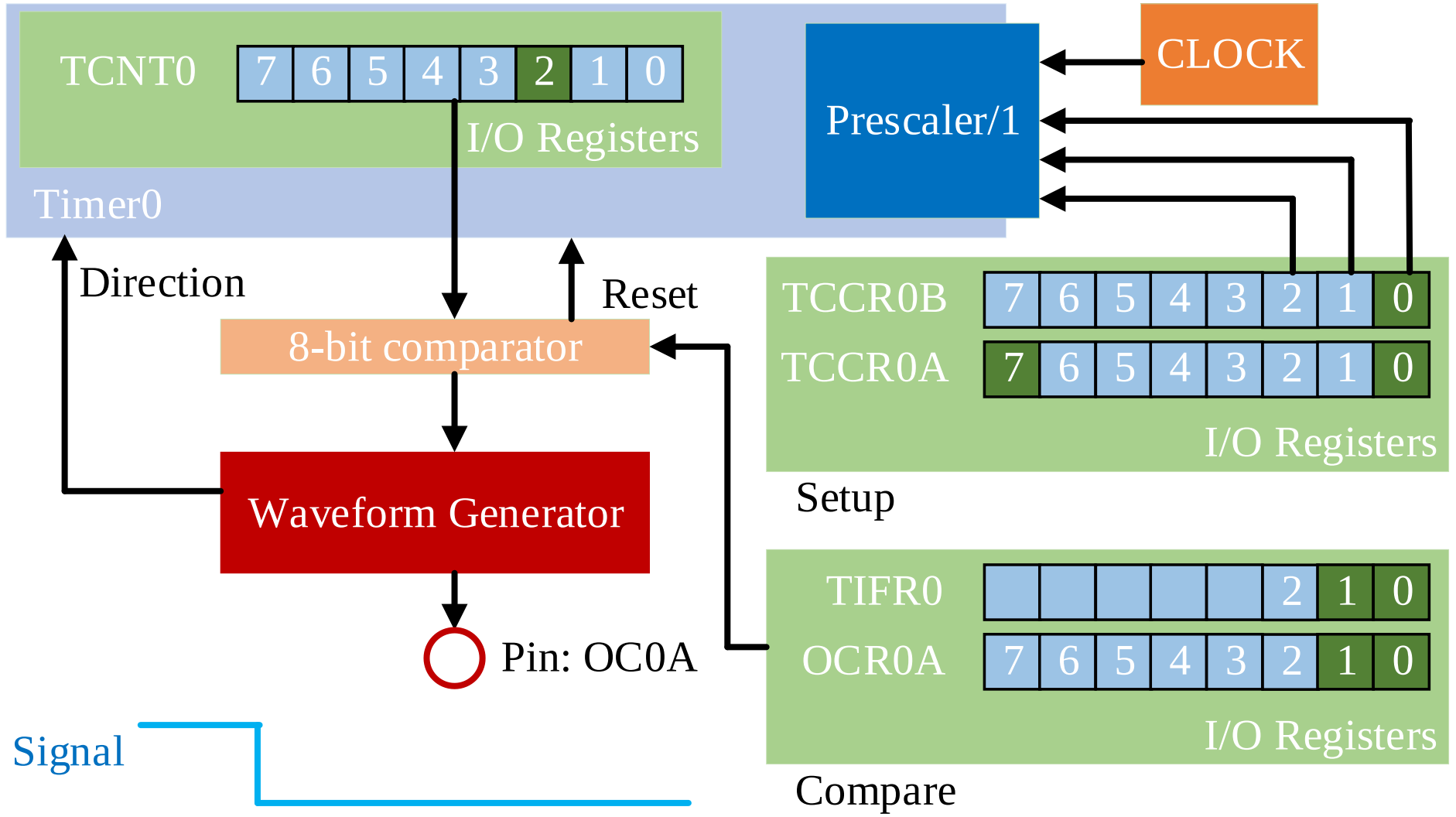


Phase Correct PWM - IX

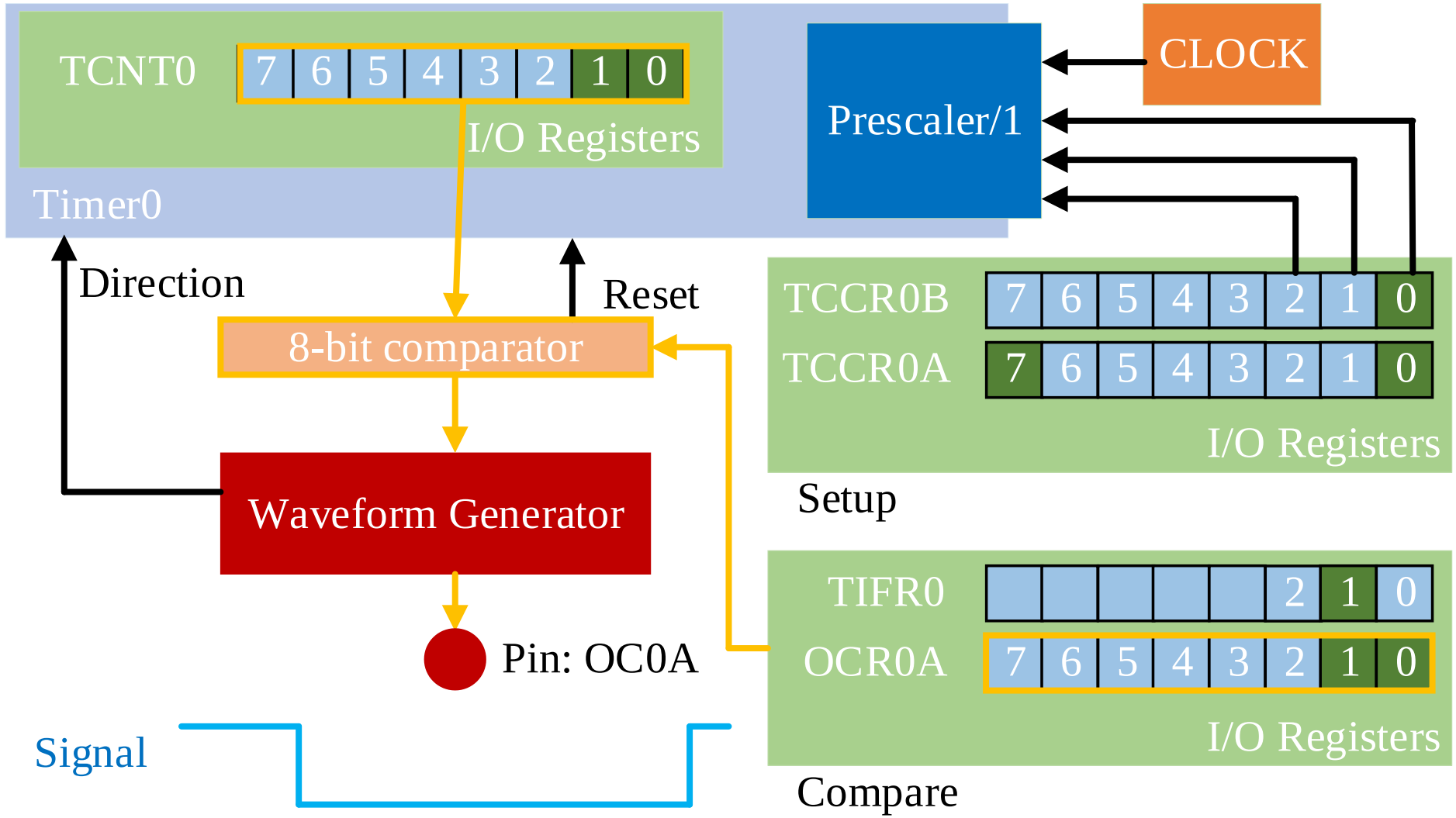
Some time later



Frequency Control - X



Phase Correct PWM - XI



Summary

- Fast PWM
  - changing duty cycl
- Phase corrected PWM